When Noisy Outcomes Reverse Criminal-Justice Fairness Conclusions

ICT 5143: Algorithms Fairness and Graphical Models Laboratory
Problem 7; Team 1

Abner Stan Fernandez (261090830008) Monis Khursheed (261090830013)

November 7, 2026

Research question and decision context

- Context: criminal-justice risk prediction using COMPAS two-year recidivism data.
- $\tilde{Y} = 1$: recorded two-year recidivism; $D = 1$: positive high-risk classification.
- Stakeholders: defendants, judges, correctional agencies, policymakers, and affected communities.
- **Question:** How sensitive are race- and sex-based fairness conclusions to group-dependent errors in recorded outcomes?
- **Hypothesis:** Feasible recording-error assumptions can reverse fairness rankings of unmitigated and mitigated classifiers; no reversals in the tested regions would refute this hypothesis there.

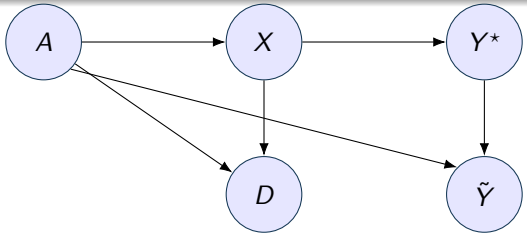
Closest work and proposed contribution

- **Closest work:** Angwin et al. (2016), *Machine Bias*, COMPAS audit; Chouldechova (2017), fairness incompatibility in recidivism instruments.
- **Open issue studied:** Fairness evaluated against recorded outcomes may differ from fairness against hypothetical latent outcomes.
- **Contribution:** Reproduced COMPAS fairness audit plus a Bayesian measurement model with configurable race-by-sex outcome-recording errors.
- **Test:** Sweep feasible error rates, compare equalized-odds rankings, validate inversion through simulations and zero-noise ablation.

Data and evaluation population

- **Data:** ProPublica public COMPAS two-year recidivism dataset (2016); published screening, charge, and quality restrictions yield $n = 6{,}172$.
- **Person-disjoint split:** training 3,703, validation 1,234, test 1,235 (60/20/20).
- **Groups:** Black–White, Female–Male, and four race-by-sex intersections.
- **Predictors:** age, prior and juvenile histories, charge degree; future outcomes excluded. Protected attributes excluded from unmitigated LR/GBT inputs.
- **Test support:** Female 235 (82 recorded positives), Male 1,000 (479 positives); sparse intersections such as Asian/Male ($n = 5$) are exploratory.

Graphical model and inference



$$p(V_1, \ldots, V_m) = \prod_{j=1}^{m} p(V_j \mid \mathrm{pa}(V_j))$$

- Query: $p(Y^\star = 1 \mid D = d, A = a; \alpha_a, \beta_a)$ via discrete variable elimination and measurement-matrix inversion.
- Assumption: $\tilde{Y} \perp (X, D) \mid (Y^\star, A)$; α_a, β_a are group-specific recording-error rates.
- Validation: forward reconstruction, feasibility checks (-10^{-10} tolerance), and numerical agreement within 10^{-12}.

Fairness criterion and proposed method

- **Equalized-odds gap:** $\Delta_{\mathrm{EO}} = \max\{|\mathrm{TPR}_a - \mathrm{TPR}_b|, |\mathrm{FPR}_a - \mathrm{FPR}_b|\}$ for two groups; lower is better.
- **Utility:** balanced accuracy and AUROC; no acceptable utility-loss threshold was prespecified.
- **Method:** compare LR with group-aware equalized-odds post-processing under hypothetical group-specific outcome-recording errors; retain only feasible inferred probabilities.
- **Protected attribute:** excluded from base LR/GBT predictors; required for group-aware EO post-processing at decision time.
- **Diagnostics:** group confusion matrices, calibration, reversal frequency, log loss and Brier score.

Experimental protocol

- **Predictive baselines:** LR, gradient-boosted trees (GBT), discrete naive Bayesian network (BN), released COMPAS scores.
- **Fairness baselines:** unmitigated LR, training-set reweighing, and EO post-processing.
- **Controls:** shared eligible features and split; training-only preprocessing; validation-only tuning and threshold selection; frozen held-out test set.
- **Uncertainty:** 1,000 paired person-level bootstrap resamples, percentile 95% intervals; randomized EO decisions sampled at seeds 0–9, with analytic expected decisions for primary estimates.
- **Stress tests:** zero/equal noise, altered noise ranges, thresholds, age adjustment, cohort windows, intersections and covariate-dependent-error simulations.

Main result and uncertainty

Method	Balanced accuracy ↑	Sex EO gap ↓	Log loss / Brier ↓
LR baseline	0.6550	0.1740	0.6128 / 0.2120
EO-sex post-processing	0.6521	0.0291	0.6128 / 0.2120
LR, zero-noise ablation	0.6550	0.1740	0.6128 / 0.2120

- Balanced-accuracy difference (EO-sex minus LR): -0.0029; paired 95% CI $[-0.0111, 0.0064]$.
- Observed fairness-gap difference (EO-sex minus LR): -0.1449; paired 95% CI $[-0.2284, 0.0051]$ (includes zero).
- **Sensitivity result:** fairness ranking reverses in 2,068/6,561 feasible sex-specific recording-error scenarios (31.52%).
- Log loss/Brier are for the unchanged underlying LR score, not EO randomized decision probabilities.

Ablation, robustness and failure case

- **Zero-noise ablation:** recovers observed-label LR fairness gap 0.1740 and balanced accuracy 0.6550.
- **Mechanism:** unequal group-specific recording errors change inferred latent decision–outcome cells and hence latent TPR/FPR gaps.
- **Failure case:** race-based LR versus EO-race has 0/6,561 reversals in the primary sensitivity grid.
- **Assumption stress test:** simulations with covariate-dependent recording errors show sensitivity to violations of conditional independence.
- **Interpretation:** 31.52% is a fraction of tested hypothetical scenarios, not a probability of real-world reversal; the bootstrap interval quantifies sampling uncertainty in the observed comparison.

Conclusion and reproducibility

- **Conclusion:** Observed-label fairness rankings need not be robust to hypothetical outcome-recording error; no claim that either classifier is truly fairer.
- **Limits/ethics:** latent outcomes and error rates are unidentified; small subgroups and historical cohort restrictions limit generalization; overinterpretation may harm defendants and communities.
- **Reproduce:** `python -m compas_fairness.run -config configs/full.json -stages models,sensitivity`.
- **Repository:** `github.com/RookieCoders/afgml-compas-criminal-dataset.git`; commit `ae3c268d441286c2847be7d7c48bbbd58dc66be3`.
- **Student 1:** Executed full research pipeline and verified audit, models, sensitivity, simulation, bootstrap, robustness, reporting and pipeline checks.
- **Student 2:** Independently verified fairness metrics and measurement-error sensitivity; reviewed Student 1's predictive-model evaluation and leakage controls (as reported in the source PDF).

Backup: mandatory evidence and references

- Evidence: `docs/REQUIREMENTS_MATRIX.md`, `results/`; full results and sensitivity analyses in the project repository.
- Angwin, J., et al. (2016). "Machine Bias." *ProPublica*. `https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing`
- Chouldechova, A. (2017). "Fair Prediction with Disparate Impact: A Study of Bias in Recidivism Prediction Instruments." *Big Data*, 5(2), 153–163. DOI: 10.1089/big.2016.0047.
- ProPublica (2016). *COMPAS Recidivism Risk Score Data and Analysis*. `https://github.com/propublica/compas-analysis`

Backup: inference and statistical validation

- Measurement model: $\alpha_a = P(\tilde{Y} = 1 \mid Y^\star = 0, A = a)$; $\beta_a = P(\tilde{Y} = 0 \mid Y^\star = 1, A = a)$.
- Conditional independence: $\tilde{Y} \perp (X, D) \mid (Y^\star, A)$; inversion requires feasible reconstructed probability cells.
- Primary sex-specific grid: four rates in $\{0, 0.025, \ldots, 0.20\}$, giving $9^4 = 6{,}561$ scenarios; 2,068 reversals.
- Test sex support: Female $n = 235$ (153 recorded negatives, 82 positives); Male $n = 1{,}000$ (521 negatives, 479 positives).
- Paired 1,000-person-bootstrap interval for fairness-gap difference: $[-0.2284, 0.0051]$; exploratory comparisons without multiplicity correction.
- Independent posterior checks agree within 10^{-12}; feasibility checks reject materially negative cells below -10^{-10}.

Backup: demonstration and individual defence

Regenerate headline result (with environment and data prepared):

```
python -m compas_fairness.run \
  --config configs/full.json \
  --stages models,sensitivity
```

Inspect saved summary:

```
python -c "import pandas as p; s=p.read_csv(
 'results/sensitivity_summary.csv'); print(s[
 (s.axis=='sex') & (s.region=='primary') &
 (s.left=='LR') & (s.right=='EO_sex')][
 ['feasible','reversed','reversal_fraction']])"
```

Expected: feasible 6,561; reversed 2,068; fraction 0.315196.

Defence: Explain measurement inversion, EO-gap construction, hypothetical versus observed outcomes, and failure under covariate-dependent errors.